

MDM: Electronics and Computer Engineering
Course: Embedded System and Application.

A Real World Case Study On

ATM MACHINE

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ATM Machine as an Embedded Application

1. Introduction

In today's modern world, embedded systems play a major role in almost every electronic device that we use. One of the best real-life examples is the **Automated Teller Machine (ATM)**. An ATM is an electronic banking outlet that allows customers to perform financial transactions such as withdrawing cash, checking account balance, transferring funds, and even changing PIN numbers without visiting a bank branch.

It works 24×7 and is designed to perform only specific tasks efficiently, which makes it a perfect example of an **embedded system application**.

2. What is an Embedded System in ATM Machine?

An **embedded system** is a combination of **hardware and software** designed to perform a **dedicated function** or a specific group of tasks. It usually works in **real-time** and is built around a **microcontroller or microprocessor**.

In the case of an ATM, all its operations like reading the card, verifying the PIN, communicating with the bank's server, and dispensing cash are handled by an embedded controller inside the machine.



3. Need for Embedded Systems in ATM

An ATM must handle thousands of transactions daily without error, ensure data security, and respond to user commands instantly. For this, the system requires:

- **Real-time processing**

- **High reliability**
- **Data security and encryption**
- **Continuous operation (24×7)**

Embedded systems provide all these features, which is why they are ideal for ATM operations.

4. Hardware Components of ATM

The ATM has several hardware components, each controlled by the embedded processor:

- **Microcontroller/Processor:** Acts as the brain of the system. It processes user inputs and controls all hardware devices.
 - **Card Reader:** Reads data stored in the magnetic stripe or chip of the ATM card.
 - **Keypad:** Allows users to enter their PIN and make menu selections.
 - **Display Screen (LCD/LED):** Shows instructions, messages, and transaction details to the user.
 - **Cash Dispenser Unit:** Mechanically counts and delivers the correct amount of money.
 - **Printer:** Prints transaction receipts.
 - **Sensors and Switches:** Detect events like card insertion, cash status, and door closure.
 - **Network Interface:** Connects the ATM to the bank's main server through secure communication.
 - **Security Camera:** Records all user activities for safety and monitoring.
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5. Software Components of ATM

The software part of the ATM is equally important as it controls how the hardware works together.

- **Operating System (RTOS):** Ensures smooth and real-time operation of multiple tasks.
- **User Interface Program:** Displays messages and menus to the customer.
- **Security Software:** Encrypts user data (especially PIN and transaction info) to prevent hacking.

- **Communication Module:** Handles network communication between ATM and bank server.
 - **Error Handling and Maintenance Software:** Detects faults like paper jams, low cash, or connection failure and reports them.
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Main Embedded Components in ATM

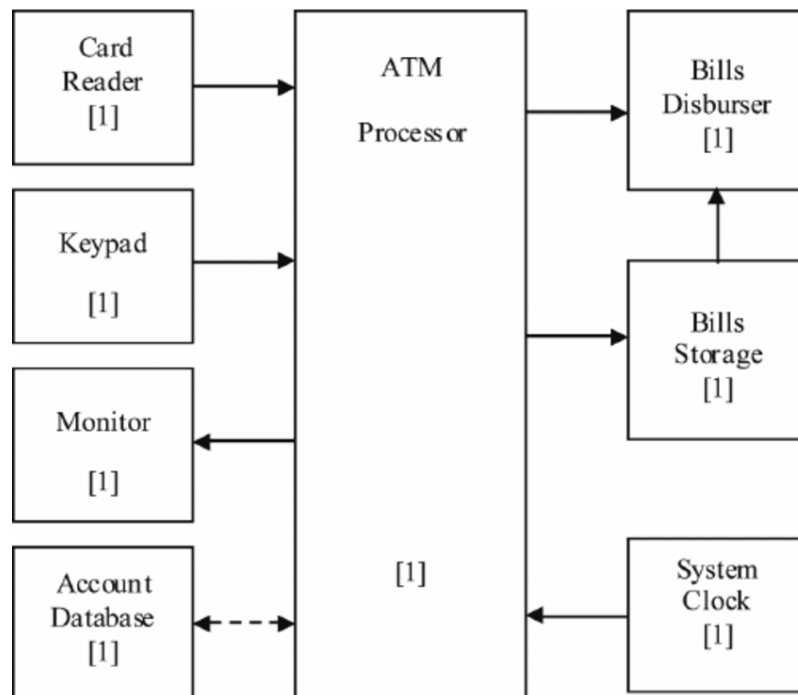
1. Microcontroller / Embedded Processor
 - It is the heart of the ATM.
 - Controls all input/output operations like card reading, keypad entry, cash dispensing, and printing.
 - Common types:
 - ARM-based processors (like ARM Cortex series)
 - Intel 8051 or advanced 32-bit microcontrollers
 2. Memory Units (RAM, ROM, Flash)
 - ROM: Stores the ATM's operating software.
 - RAM: Used for temporary data (like transaction info).
 - Flash memory: Stores system updates and logs.
 3. I/O Interfaces
 - Connect the microcontroller with components such as keypad, display, printer, and network module.
 4. Communication Module
 - Handles data exchange between ATM and bank's central server via secure network protocols (TCP/IP, SSL).
 5. RTOS (Real-Time Operating System)
 - Manages multiple tasks like reading card data, verifying PIN, and dispensing cash simultaneously.
 - Ensures real-time operation without delay.
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6. Understanding of System Architecture

An ATM system is built using both hardware and software components which work together under the control of an embedded microcontroller.

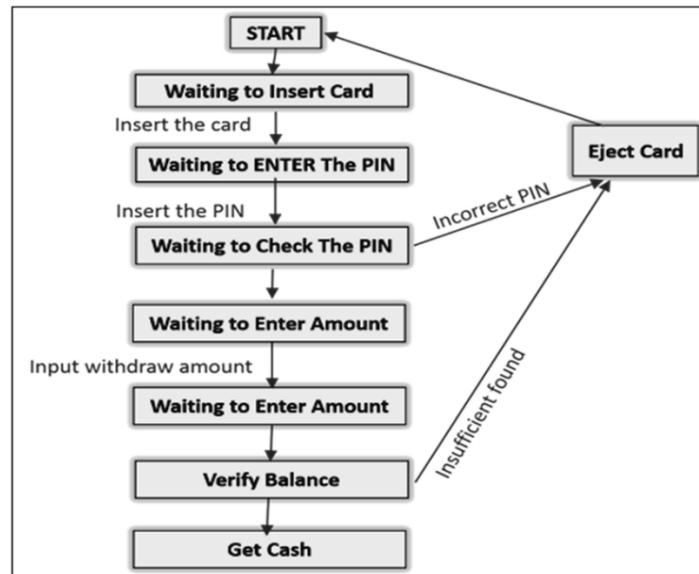
The main function of the system is to communicate with the bank's central server, verify user identity, and perform secure money transactions.

Block Diagram:



Working Principle

1. The user inserts the card into the reader.
2. The microcontroller reads account data from the card's magnetic strip or chip.
3. The user enters the PIN, which is encrypted and sent to the bank server through the network interface.
4. The server authenticates the user and sends back approval.
5. The user selects the desired transaction (e.g., cash withdrawal).
6. The microcontroller sends a command to the **cash dispenser** to release the required amount.
7. The **printer** prints a receipt and the **display** shows the transaction summary.
8. Finally, the card is ejected and the session ends.



7. Evaluation of ATM as an Embedded System

Parameter	Description
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Functionality	Performs specific banking operations securely and automatically.
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Performance	Real-time response with fast processing speed.
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Reliability	Works continuously with minimal errors or downtime.
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Security	Ensures data encryption, user authentication, and camera monitoring.
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User Interface	Easy to use with simple menus and clear instructions.
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Maintenance	Requires periodic cash refill, cleaning, and software updates.
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8. Advantages

- Provides 24-hour banking service without human assistance.
 - Saves time for both customers and banks.
 - Reduces long queues in banks.
 - Simple to use and accessible in multiple locations.
 - Minimizes the need for human error in transactions.
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9. Limitations

- Vulnerable to card skimming and hacking if not properly secured.
 - Requires continuous network connection and power supply.
 - Maintenance cost is high.
 - Sometimes issues like card jam or network failure can cause inconvenience.
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10. Real-World Impact

ATMs have completely changed the way people handle banking transactions. They brought **automation** into the banking sector and improved **financial accessibility** even in rural areas. With the rise of **smart ATMs**, users can now perform mini statement checks, mobile recharges, and even deposit cash — showing how embedded systems continue to evolve and make life easier.

11. Conclusion

The **ATM Machine** is one of the best real-world examples of an **embedded system application**. It combines electronic hardware, secure software, and networking to provide safe, reliable, and user-friendly banking services.

This case study shows how embedded systems simplify complex human tasks and play a vital role in automating everyday processes around us.

14. References

- Wikipedia: *Automated Teller Machine (ATM)* – https://en.wikipedia.org/wiki/Automated_teller_machine
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